New comfort index during combined conditions of moderate low ambient temperature and traffic noise

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Abstract

This study's aim is to propose a new comfort index for indicating the combined effect of cold and noise stress on the human state of mind. Twenty-two male students were exposed to 20 combined conditions involving four operative temperature levels and five noise levels. The subjects reported their sensations regarding each combined condition. The results show that the auditory condition significantly affected the hot sensation as well as the noise sensation, and that the thermal condition also significantly affected the noise sensation. Both temperature and noise affected obviously the universal comfort and discomfort sensations. Consequently, two kinds of equi-comfort charts were derived. One of the charts, which represents the equal universal comfort sensation derived from the combination of thermal and auditory comfort sensation, demonstrates the exclusivity of the combined effects. The other chart indicates temperature and noise levels in order to quantitatively evaluate the combined effect of cold and noisy conditions based on the experimental results. This chart can reasonably predict human comfort sensations within this experimental condition.

Keywords

Temperature; Noise; Specific comfort; Universal comfort; Equi-comfort chart; Exclusivity

1. Introduction

The purpose of studies on the combined effects of environmental factors is to clarify not only the individual effects of each factor but also their interaction, which cannot be obtained in studies on specific factors alone. The present experimental study focuses on the combined effect of lower ambient temperature and noise level on human psychological response and aims to propose a new comfort index for their combined effects. Our previous study [1] proposed a comfort index for combined hot and noisy conditions, but did not focus on cooler environments. Viteles and Smith [2], Grether et al. [3, 4], Poulton et al. [5], and Bell [6] investigated the combined effects of heat and noise on task performance, but did not investigate cooler conditions. Yamazaki et al. [7] and Mochizuki et al. [8] also took only one cool condition into account in addition to one neutral and one warm condition. Horie et al. [9] and Sakurai et al. [10] quantified non-specific discomfort in combined cool, noisy, and

bright conditions, but did not investigate each of these sensations individually or the specific comfort evaluation of each factor (i.e., thermal comfort/discomfort, the auditory, etc.). It is of great importance to measure these specific variables as well as non-specific ones in order to determine the role of specific sensations in establishing broader sensations such as universal comfort. Matsubara [11] investigated impressions of both individual discomfort and universal discomfort, but did not clarify their combined effects quantitatively. The present study measured and discussed the human state of mind under the combined conditions of a wide range of lower ambient temperature and traffic noise levels.

2. Materials and methods

2.1 Subjects

Twenty-two of the participants in our previous experiments [1] were adopted as the subjects of this study. They were healthy Japanese male students with normal hearing, ranging in age from 19 to 26 years old (Mean (SD) = 22.1 (1.9)). Prior to the experiment, each subject gave his informed consent, but was not informed of the concrete experimental conditions in order to prevent bias or prejudice concerning each condition. All of the subjects participated voluntarily in all sessions, and were compensated for their participation. In the experimental chamber they wore only cotton undershorts and remained in a sedentary posture.

2.2 Experimental conditions

The pre-test room was kept at an operative temperature of 27°C that might create a thermally neutral effect and at an equivalent sound pressure level of 53.9dB(A) from air-conditioner. In the test room, four operative temperature levels (19, 22, 25, and 28°C) at each of five equivalent sound levels (46.6dB(A): air-conditioning noise; 58.5, 72.9, 79.9, and 95.5dB(A): traffic noise) constituted 20 experimental conditions. Relative humidity and air velocity were approximately 30−60% and less than 0.15m/s, respectively, at the occupied zone in the test room and the pre-test room. Table 1 shows the actual thermal conditions.

2.3 Experimental facilities

The experimental program was conducted at the experiment chamber in the Nagoya Institute of Technology. Figure 1 shows the floor plan of the experimental rooms. Two rooms were constructed of steel frames and polystyrene foams. The dimensions of the pre-test room were 2.4 m × 2.7 m with a ceiling height of 2.4 m. Those of the test room were 3.6 m in width, 3.0 m long, and 2.4 m in height. The interior surfaces of each room were covered by a gray-colored (N8.5) curtain. It was possible to separately control the air temperatures of each room by means of the packaged air-conditioner. Traffic noise was recorded on the bridge over a highway in a Nagoya suburb and was presented using a digital audio tape recorder (SONY

TCD-D10 PRO II). Noise levels were adjusted using the volume of the amplifier (AIWA S-A22).

2.4 Measurement

The ambient air temperatures and wet-bulb temperatures in the pre-test and test rooms were measured using a ventilating psychrometer positioned 700 mm high from the floor, and the globe temperature was measured using a globe thermometer. Along with these temperatures, the surface temperatures of the walls, floor, and ceiling and the vertical distribution of the air temperature were continuously measured and monitored by means of 0.3mm T-typed thermocouples. Mean radiant temperatures for determining the operative temperature [12] were calculated using the surface temperature of the wall, floor, and ceiling and the angle factor between a subject (as the 500 mm-sized cube model [13]) and each surface. Twenty linear unipolar scales were adopted for the psychological measurements as shown in Figure 2. Each Japanese term is located on the right side of the line that indicates the highest point, and its negative term is appended to the left for the subjects' easy comprehension. The subjects were instructed to put a mark on the line of which the position described their sensations in the environment. A numerical value from 0 to 100 was assigned to each response on the scale.

2.5 Procedure

The time schedule of the experiment is shown in Figure 3. All experiments were conducted in the morning (9:00−12:30) or afternoon (13:30−19:00) from 2 to 18 February 1999. The experiments were not conducted an hour after breakfast and lunch or a coffee break in order to ensure the subjects' metabolic stability. After a half-hour of thermal adaptation in the pre-test room, the subjects walked into the test room and sat on the chair. They registered their impression of the experimental environment on a ballot after being presenting the noise stimuli for 2 min. After 7 min, they repeated the ratings for the same exposure condition. At the 19°C condition, however, the second set of ratings was not conducted in order to protect the subjects from excessive physical and mental stress. This paper analyzed the second set of ratings, which were conducted at a more steady state, adding to the ratings for the 19°C condition, because the mean votes of the first set of the ratings were almost identical to those of the second set. ANOVAs for a two-factor model of temperature and noise were used for each indication of sensation in order to statistically investigate the effect of each environmental factor and their interaction [14].

3. Results

3.1 The relationship between the experimental conditions and thermal sensations

Fig. 4(a)−(f) show the relationship between the operative temperature and the mean values

of thermal sensation votes. With the increase of the operative temperature, the cold, cool, and thermal discomfort sensation votes decreased while the hot, warm, and thermal comfort sensation votes rose to a higher level. Table 2 shows the results of two-way ANOVAs for each sensation vote. The cold sensation votes were significantly associated with the main effect of temperature ($F=78.81$, $p<0.01$), but not with the main effect of noise ($F=1.19$, n.s.) and with their interaction ($F=0.45$, n.s). For the hot sensation votes, ANOVA indicated that these votes were significantly associated with the main effect of temperature ($F=13.09$, $p<0.01$) and with their interaction ($F=1.91$, $p<0.05$), but not with the main effect of noise ($F=0.52$, n.s.). The multiple comparison procedure for the interaction indicated that there was a significant difference between 28°C−46.6dB (A) and 28°C−79.9dB (A) conditions only (Tukey's HSD test, $t=6.00$, $p<0.05$). The thermal comfort sensation votes were not only significantly associated with the main effects of temperature ($F=58.71$, $p<0.01$) but noise as well ($F=10.69$, $p<0.01$). The thermal discomfort sensation votes were also associated significantly with the main effects of temperature ($F=44.56$, $p<0.01$) and noise ($F=10.17$, $p<0.01$). Table 3(a) and (b) show the mean values of thermal comfort and discomfort votes, respectively, in addition to each result of Tukey's HSD test for the main effects of noise. As the noise level increased, the mean values of thermal sensation votes decreased, and vice versa with regard to the thermal discomfort sensation votes. As shown in Tables 3(a) and (b), five and four pairwise differences were significant, respectively.

3.2 The relationship between the experimental conditions and auditory sensations

Fig. 5(a)−(e) show the relationship between the noise level and the mean values of auditory sensation votes. As the noise level increased, the subjects registered their impressions of more noise, more annoyance, and more auditory discomfort, and vice versa regarding the quiet and auditory comfort sensation votes, with little respect to the operative temperature. As shown in Table 2, the noisy sensation votes were significantly associated with the main effect of noise ($F=148.40$, $p<0.01$) and insignificantly with temperature ($F=2.26$, $p=0.09$), but not with the interaction ($F=1.18$, n.s.). The multiple comparison procedure for temperature indicated an insignificant difference between the 19 and 25°C conditions (Tukey's HSD test, $t=4.47$, $p=0.09$). The annoyance and quiet sensation votes indicated significant influence by the main effect of noise ($F=152.10$, $p<0.01$; $F=121.89$, $p<0.01$), but not by the main effect of temperature ($F=0.43$, n.s.; $F=0.18$, n.s.) and their interaction ($F=1.03$, n.s.; $F=0.88$, n.s.). The ANOVA of the auditory comfort sensation votes indicated significant affects of both the main effects of noise ($F=50.93$, $p<0.01$) and temperature ($F=3.35$, $p<0.05$). The auditory discomfort sensation votes showed significant influence by the main effect of noise ($F=84.23$, $p<0.01$), but not by the main effect of temperature ($F=1.92$, n.s.) and by their interaction ($F=0.83$, n.s).

3.3 The relationship between the experimental conditions and universal sensations

Fig. 6(a) and (b) show the relationship between operative temperature and the mean values of universal comfort and discomfort sensation votes, respectively. With rising operative temperatures, and decreasing noise levels, the universal comfort sensation votes increased and the universal discomfort sensation votes decreased.

3.4 The relationship between specific and universal comfort sensations

Fig. 7(a) and (b) show diagrams of the combinations of thermal and auditory comfort and discomfort sensations as lines representing equal universal comfort and discomfort sensations, respectively. These lines are interpolated by means of the spline function. When thermal and auditory comfort sensations indicate similar levels on each scale as shown in Fig. 7(a), and when thermal and auditory discomfort sensations indicate similar levels as shown in Fig. 7(b), these sensations are affected by both of the specific comfort sensations and both of the specific discomfort sensations, respectively. However, when there is an extreme difference between thermal and auditory comfort/discomfort sensation levels, universal comfort and discomfort depend on the more uncomfortable specific sensation level, not the comfortable one.

4. Discussion

4.1 The combined effects on each sensation

The results show that noise level had little effect on the cold, cool, and warm sensations, as seen in Fig. 4(a)−(d) and in the results of statistical analyses. With regard to the hot sensation, statistical significance was seen in the difference of the noise level. But as shown in Fig. 4(b), the effect of noise on the votes was very slight. Borsky et al. [15] have also shown that noise significantly but very slightly causes an increase in the subjective evaluation of thermic feeling (SETF). As well, other studies [1, 8, 11, 16, 17] found no effects of noise on thermal sensation. However, thermal comfort and discomfort sensations significantly decreased and increased with noise, and these results are reproducible as shown in previous studies [1, 17]. Thermal comfort and discomfort are closely linked to thermal sensation, respectively, and there is still an absence of firm evidence supporting noise level's lack of effect on human thermal sensation. Thus, the auditory factor should rather be considered in examinations of heat's effect on the human state of mind, especially in investigations of slight differences and fluctuations in psychological response.

The relationship between the thermal environmental factor and the auditory sensation votes were similar to that of the auditory factor and the thermal responses mentioned above. The results shown in Fig. 5(a)−(e) and ANOVAs show that the noise, annoyance, and quiet sensation votes were scarcely influenced by operative temperature, while the operative temperature showed slight effects on auditory comfort sensation votes. The thermal

environment should be taken into account in the acoustical study.

With regard to the universal comfort and discomfort sensations, both temperature and noise have obvious effects as shown in Fig. 6(a) and (b). These results can be explained as follows: The environmental stimuli inputted via each sensory organ activate primarily hot, cold, noisy, or bright sensations, and then this sensory information is integrated to determine comfort sensation, as shown in Fig. 8. That is, the specific evaluation of thermal and auditory comfort, as well as universal comfort, can involve more multisense meanings than strictly sensory evaluation, and are likely to be influenced by various factors.

4.2 Equi-comfort chart constituted with the specific comfort

The diagrams shown in Fig. 7(a) and (b) indicate that an environment that contains a single uncomfortable factor is essentially considered uncomfortable, with little respect to comfortable sensations produced by another factors. Mori [18] suggested that when one attribute was processed in parallel via plural information, the human could perceive only simple information depending on the exclusive integration principle. If thermal and auditory comfort represent in parallel one psychological attribute, that is, universal comfort, this finding of this study supports Mori's Exclusivity Principle. The same result was also derived from our previous experiments [1] conducted under hot conditions. Horie et al. [9] and Sakurai et al. [10] indicated that “discomfort” measured on a subjective scale was determined predominantly by the thermal condition when it was in the extreme range, and the effect of noise level may be masked. Yamazaki et al. [7] described that the thermal effect on subjective “work suitability” was masked when the subjects were exposed to loud noise. Hane et al. [19] noted that thermal condition did not affect universal comfort during exposure to methyl mercaptan, which smells like rotten onions. These previous studies support the findings of the present study. Thus, the exclusivity of the combined effects can be derived in common from various studies that take differing environmental factors into account and adopt the different evaluation methods.

The diagrams also represent the magnitude of the effect of the specific evaluation on the universal evaluation, and allow us to associate the findings from studies taking simple environmental factor into account with those of studies on universal evaluation.

4.3 The new comfort index constituted with thermal and auditory condition

As shown in Fig. 6(a) and (b), the scales of universal comfort and discomfort can simultaneously evaluate thermal and auditory environments. Fig. 9(a) and (b) show diagrams depicting combinations of thermal and auditory conditions as lines that represent equal universal comfort and discomfort sensations. These lines are interpolated by means of the spline function. The relationship of the universal comfort/discomfort sensation and two

physical variables on the two dimensional space are not linear. These diagrams are equi-comfort and equi-discomfort charts, respectively, composed of the operative temperature and the noise level. Horie et al. [9] and Sakurai et al. [10] have proposed the additive model for the quantitative evaluation of the combined effects of temperature, noise, and illuminance. However, this model is valid only when the universal evaluation can be expressed as the addition of the degree of each environmental stress. As mentioned above, when an extremely uncomfortable factor is present in the environment, the effect of other factors may be excluded. As well, this model can predict the three degrees of discomfort only. On the other hand, the charts shown in Fig. 9(a) and (b) can quantitatively evaluate the combined effects of cold and noisy conditions without being restricted to such additive operations and moderate ranges of each environmental stress, and can predict universal comfort and discomfort in multistep values. Consequently, these charts make it possible to reasonably predict human comfort and discomfort sensations based on the given temperature and noise level within this experimental condition.

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Table 1. Actual thermal conditions

Mean (SD)	19ºC	22ºC	25ºC	28ºC
operative temperature [ºC]	18.9 (0.4)	21.6 (0.4)	25.0 (0.4)	28.1 (0.3)
relative humidity [%]	47 (2)	46 (5)	41 (7)	36 (4)

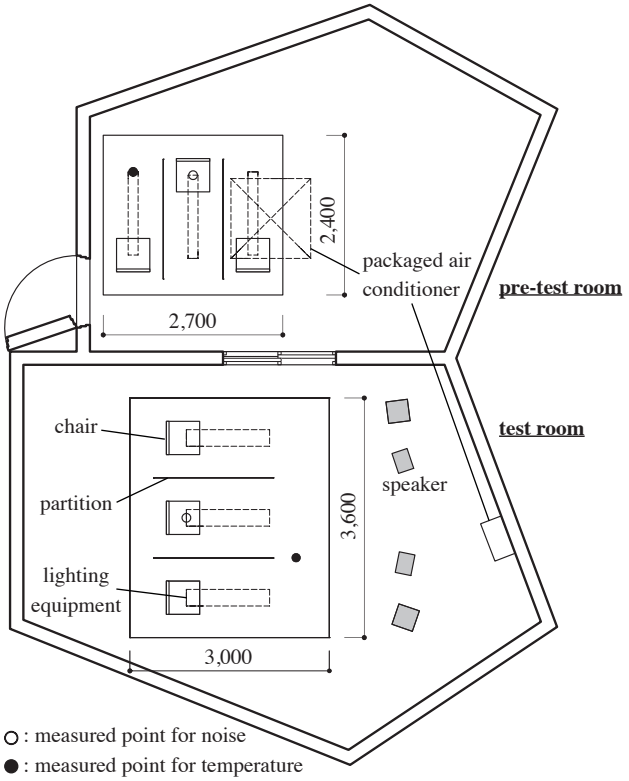


Fig. 1. Floor plan of experimental rooms

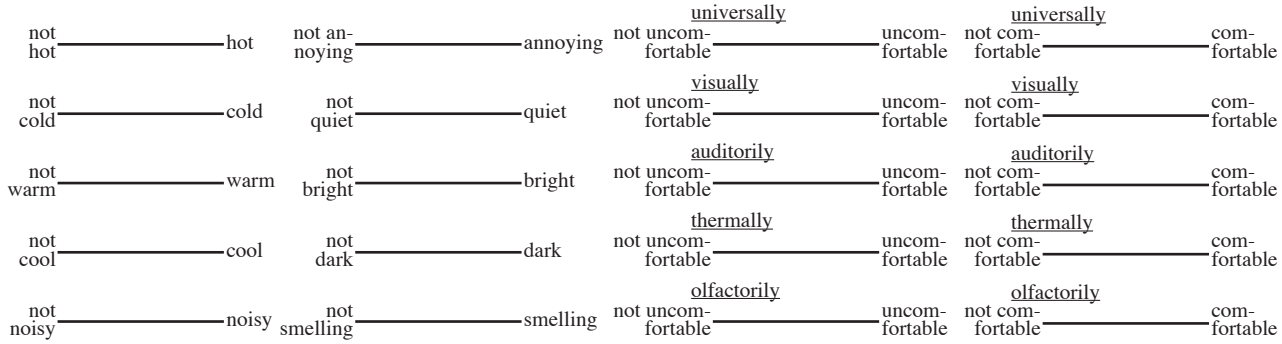


Fig. 2. Unipolar scales used for the measurements of the subject's feelings to his environment.

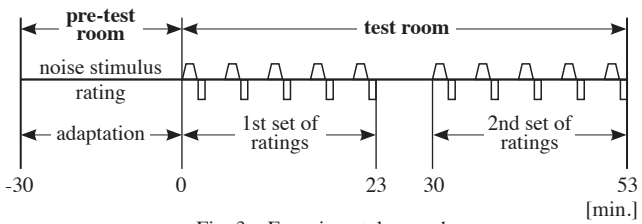
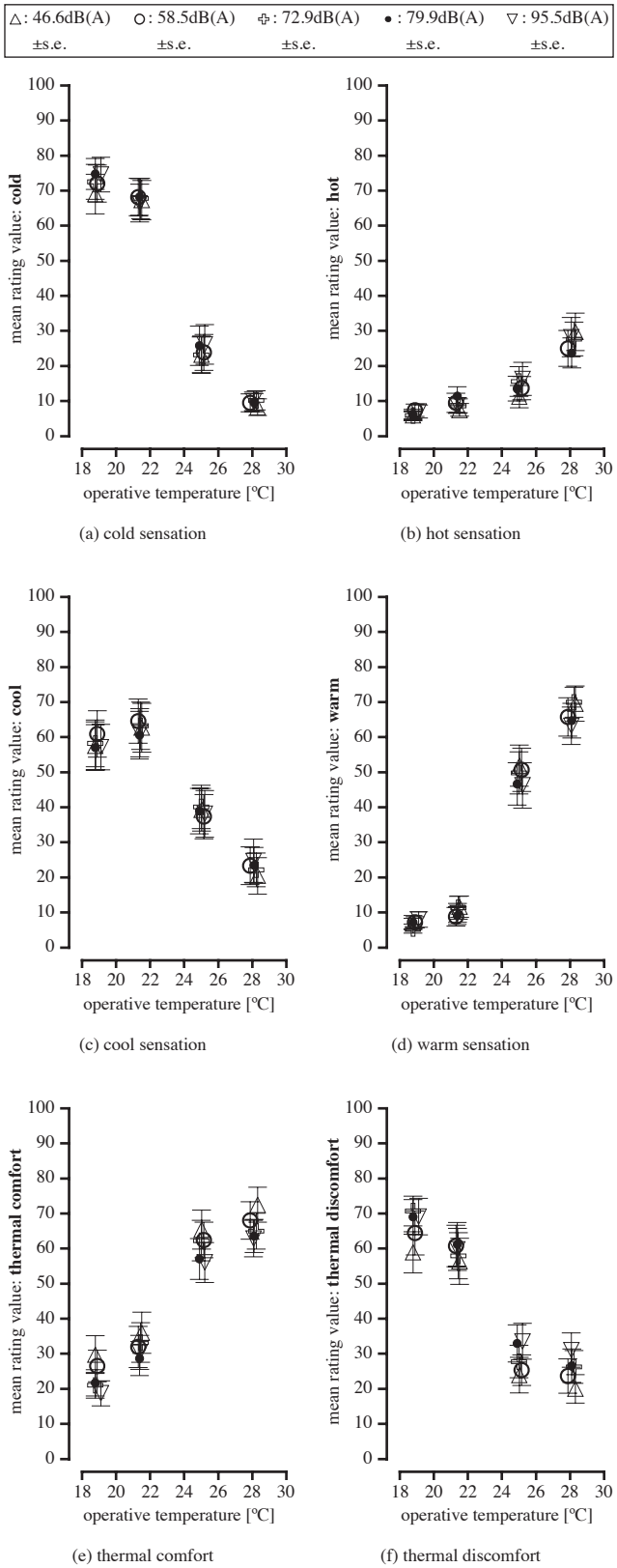


Fig. 3. Experimental procedure

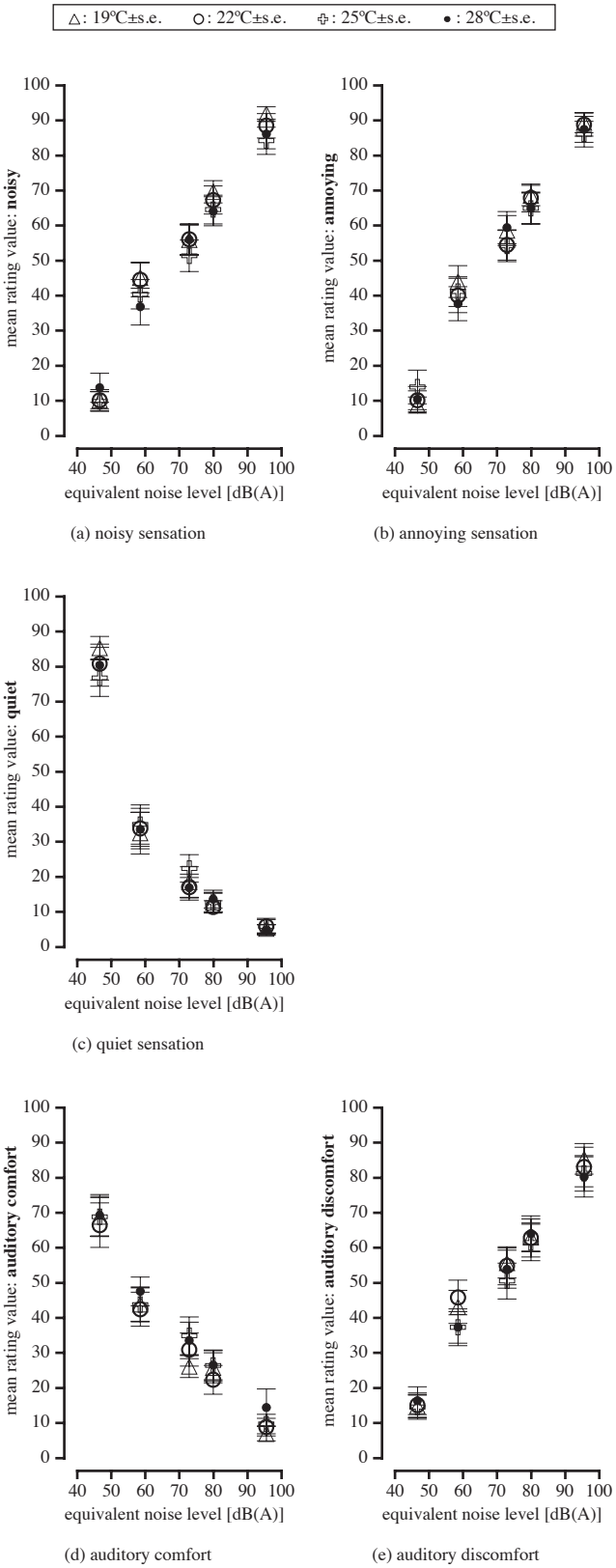


(a) cold sensation (b) hot sensation

(c) cool sensation (d) warm sensation

(e) thermal comfort (f) thermal discomfort

Fig. 4. The relationship between operative temperature and thermal sensation vote



(a) noisy sensation (b) annoying sensation

(c) quiet sensation

(d) auditory comfort (e) auditory discomfort

Fig. 5. The relationship between equivalent noise level and auditory sensation vote

Table 2. The results of two-way ANOVAs

probability	cold	hot	cool	warm	thermal comfort	thermal discomfort	noisy	annoying	quiet	auditory comfort	auditory discomfort
main effect of temperature	0.00	0.00	0.00	0.00	0.00	0.00	0.09	0.73	0.91	0.02	0.14
main effect of noise	0.32	0.72	0.76	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00
interaction of temperature and noise	0.94	0.03	0.83	0.49	0.95	0.46	0.30	0.42	0.57	0.91	0.62

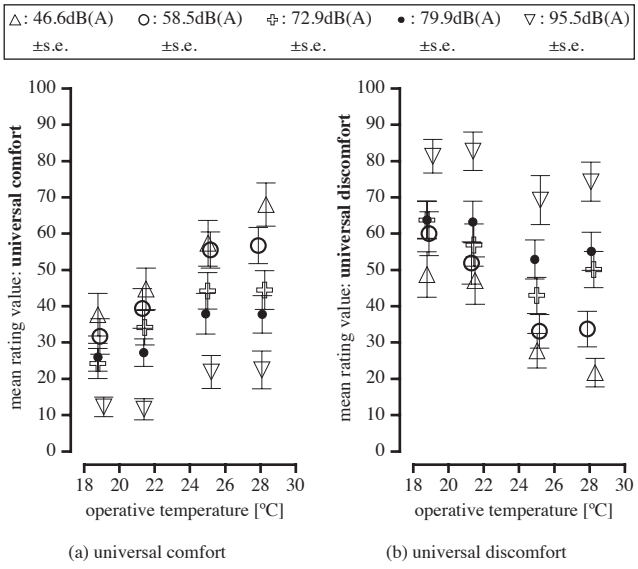
Table 3. The mean values of thermal comfort and discomfort votes, and the results of Tukey's HSD test for the main effect of noise

(a) thermal comfort

	mean	probability			
		58.5 dB(A)	72.9 dB(A)	79.9 dB(A)	95.5 dB(A)
46.6 dB(A)	50.91	0.14	0.01	0.00	0.00
58.5 dB(A)	47.23		0.75	0.04	0.01
72.9 dB(A)	45.36			0.45	0.21
79.9 dB(A)	42.75				0.99
95.5 dB(A)	42.00				

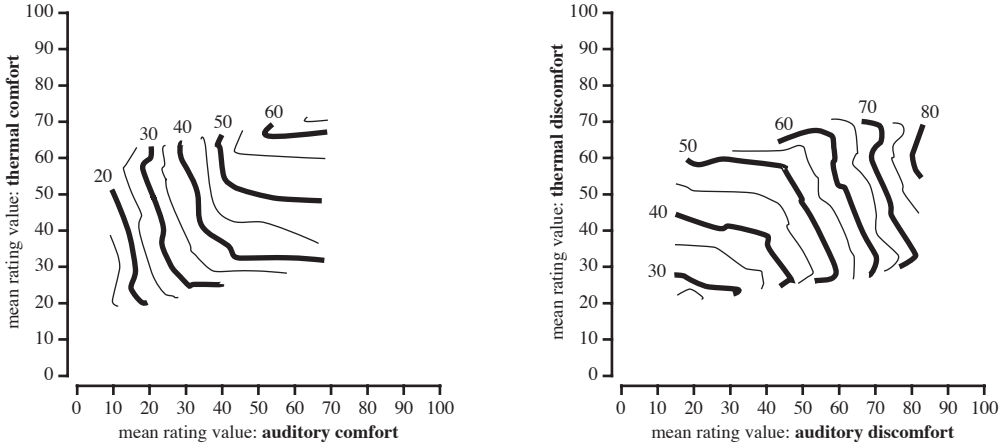
(b) thermal discomfort

	mean	probability			
		58.5 dB(A)	72.9 dB(A)	79.9 dB(A)	95.5 dB(A)
46.6 dB(A)	39.81	0.11	0.00	0.00	0.00
58.5 dB(A)	43.55		0.64	0.09	0.02
72.9 dB(A)	45.64			0.77	0.38
79.9 dB(A)	47.42				0.97
95.5 dB(A)	48.36				



(a) universal comfort (b) universal discomfort

Fig. 6. The relationship between operative temperature and universal evaluation



(a) universal comfort (b) universal discomfort

Fig. 7. Equi-comfort and equi-discomfort charts constituted with auditory and thermal evaluation

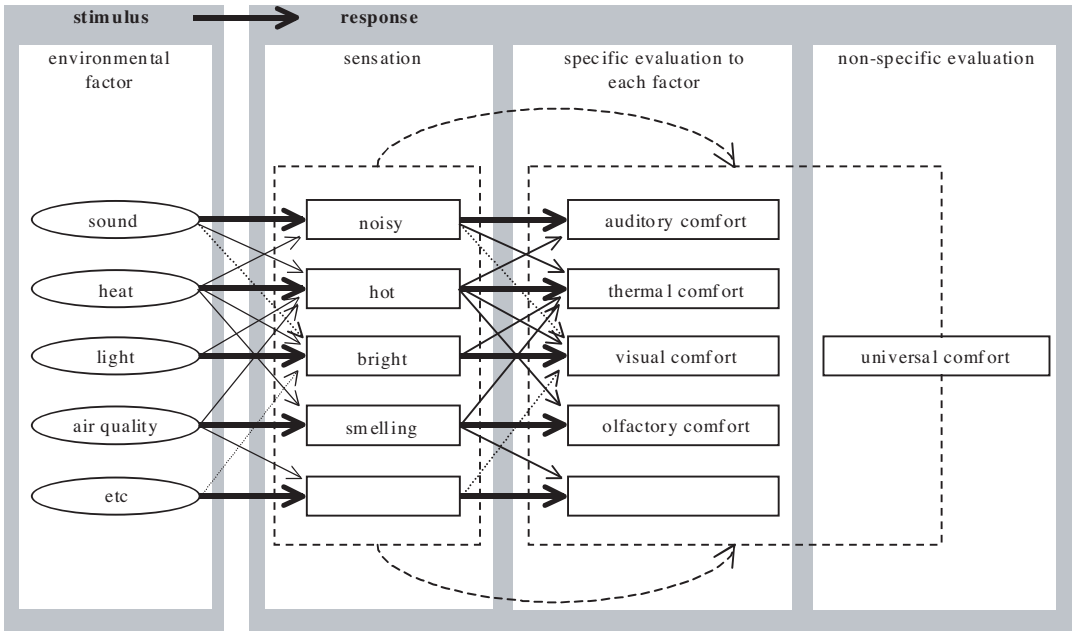
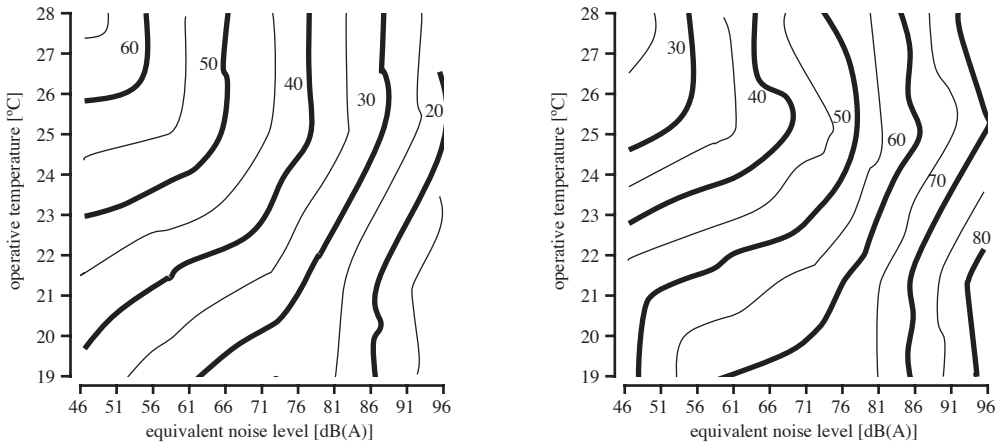


Fig. 8 Model for combined effects of environmental variables on human psychological responses



(a) universal comfort (b) universal discomfort

Fig. 9. Equi-comfort and equi-discomfort charts constituted with equivalent noise level and operative temperature